

Future Prospects in Gravitational Waves: From Testing Fundamental Physics to Instruments beyond LIGO

Brian Seymour

Caltech

Thesis Defense

May 05, 2025

Committee Members

Katerina Chatziioannou

Yanbei Chen

Saul Teukolsky

Kathryn Zurek

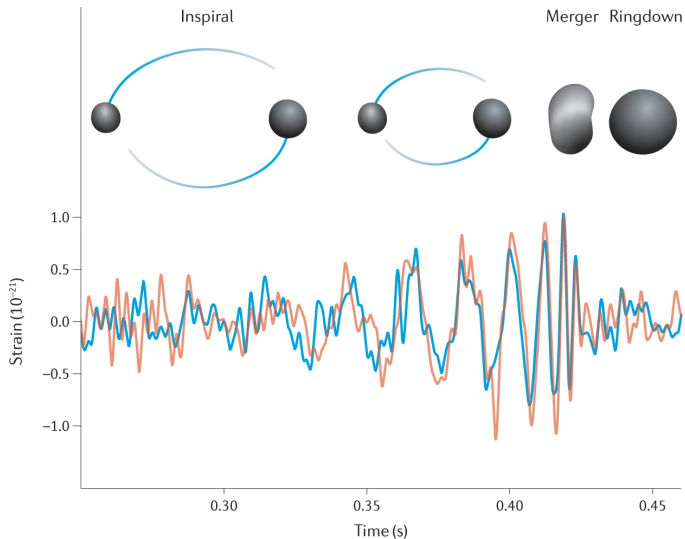
Supervisor: Yanbei Chen

Outline

- 1 Introduction
 - Science with Gravitational Waves
 - Features of GR & Beyond GR Waveforms
- 2 Part A : Modeling the Gravitational Waves in Nonviolent Nonlocality
- 3 Part B : Geometric Description of Tests of GR
 - Geometry of Waveform Deviations
 - Generic Behavior of Parameterized Tests
 - Multiparameter Tests with Singular Value Decomposition
- 4 Part C : High Frequency Gravitational Wave Detection
- 5 Conclusion

(BS: add discription of nvnl + subsection)

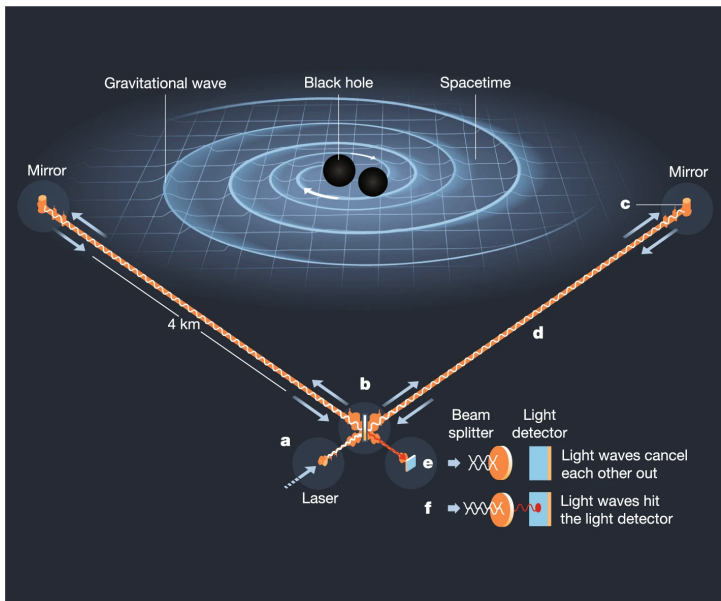
Introduction



- Image of first detection of gravitational waves in 2015.

(BS: add qscan)

[LIGO+ 2016, Bailes+ 2021]



[Miller+ 2019]

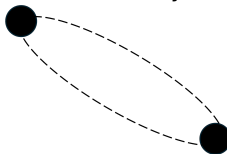
Current & Future Science Goals

- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity

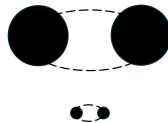
Large Asymmetric Spins



Eccentricity

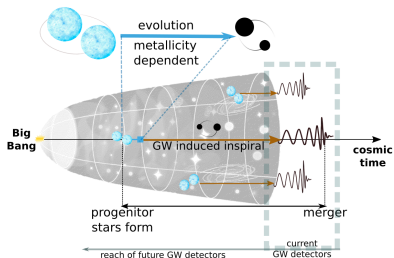


Large or Small Masses



Current & Future Science Goals

- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



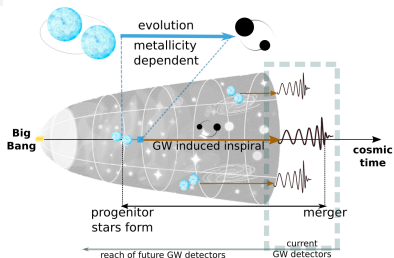
(BS:

**Astrophysics of compact objects:
formation scenarios and exotic systems)**

[Chruslinska 2022, LIGO+ 2023]

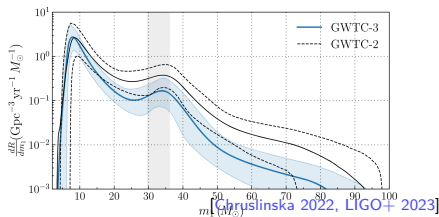
Current & Future Science Goals

- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



(BS:

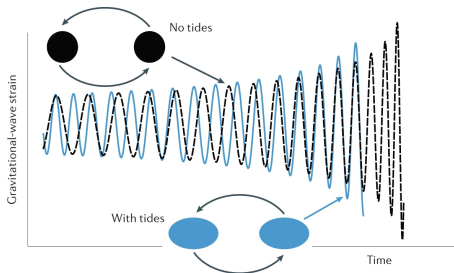
Astrophysics of compact objects:
formation scenarios and exotic systems)



[Chruslinska 2022, LIGO+ 2023]

Current & Future Science Goals

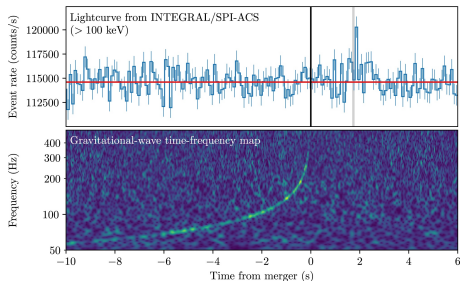
- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state**
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



[Carson 2020]

Current & Future Science Goals

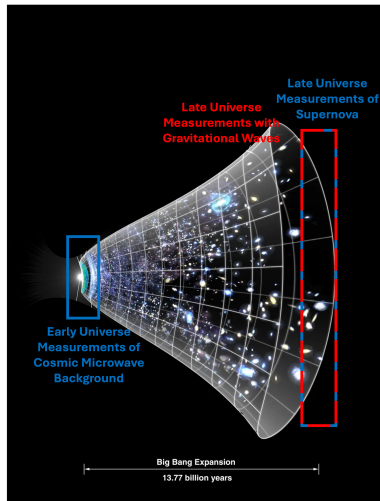
- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova**
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



[LIGO+ 2017]

Current & Future Science Goals

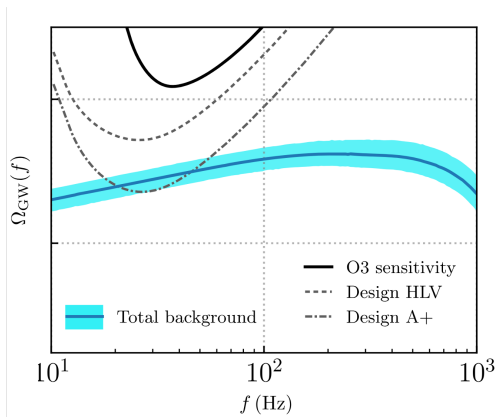
- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 **Cosmology: independently measuring expansion of universe**
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



[NASA / WMAP Science Team]

Current & Future Science Goals

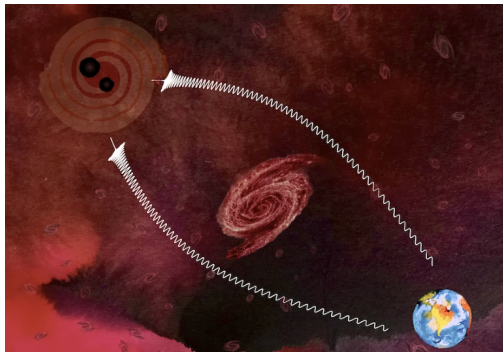
- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves**
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity



[LIGO+ 2023]

Current & Future Science Goals

- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures**
- 8 Fundamental physics: testing consistency with general relativity



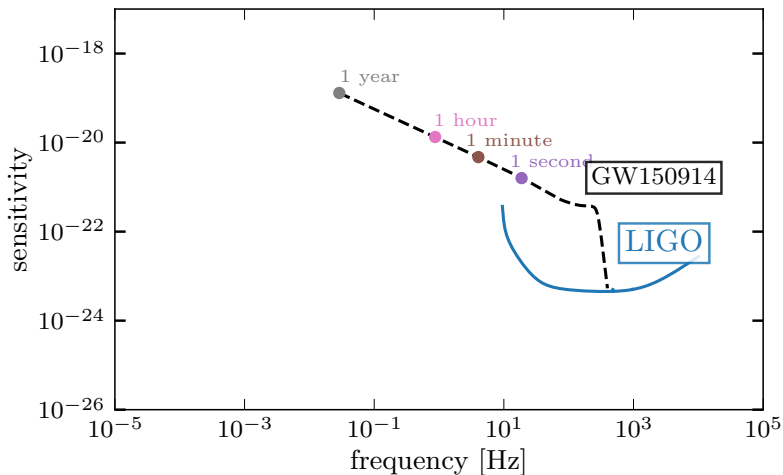
[Jana+ 2023]

Current & Future Science Goals

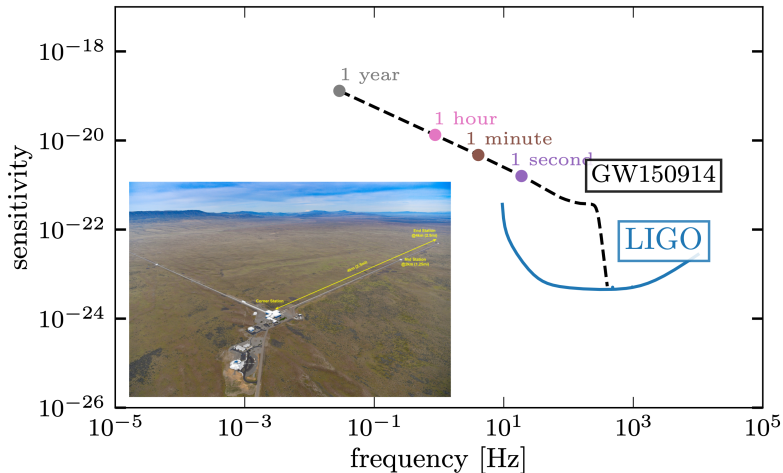
- 1 Find exotic systems: high spin, outlier masses, eccentricity ...
- 2 Rates & populations: formation scenarios, mass gaps, hierarchical mergers
- 3 Nuclear physics: tidal deformability and neutron-star equation of state
- 4 Multimessenger astrophysics: electromagnetic counterparts and kilonova
- 5 Cosmology: independently measuring expansion of universe
- 6 Stochastic background: unresolved background of gravitational waves
- 7 Lensing signatures
- 8 Fundamental physics: testing consistency with general relativity**

(BS: must add figure here and fix)

Current & Future Detectors

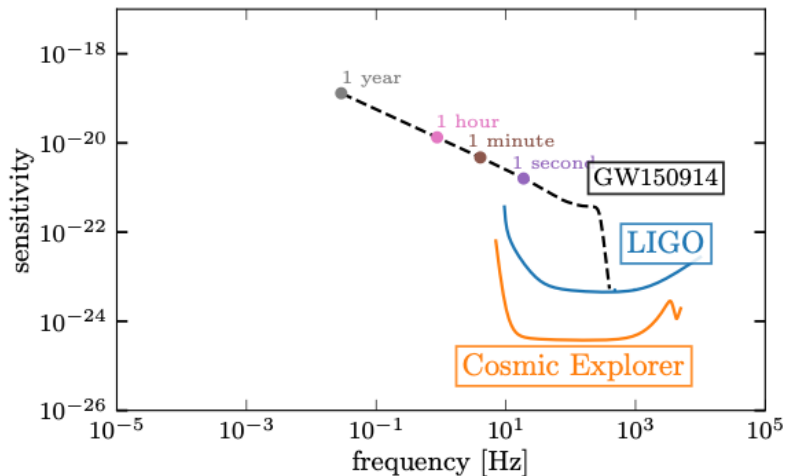


Current & Future Detectors

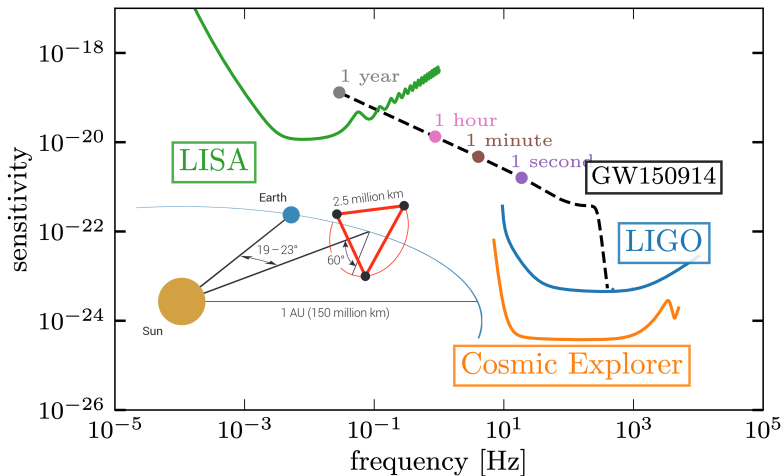


[LIGO Lab]

Current & Future Detectors

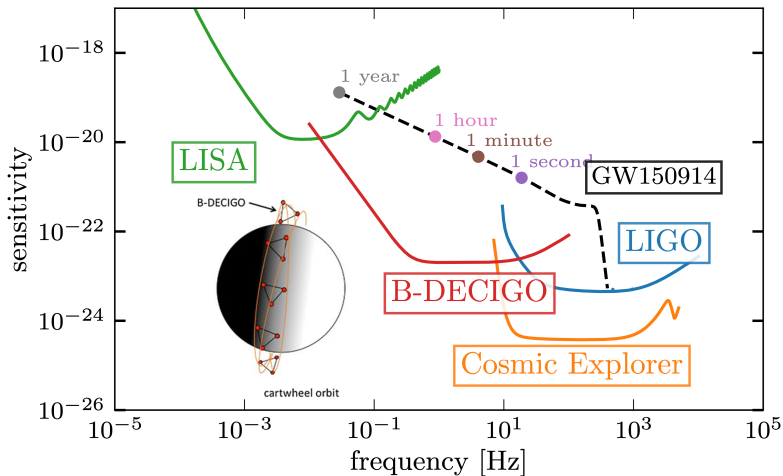


Current & Future Detectors



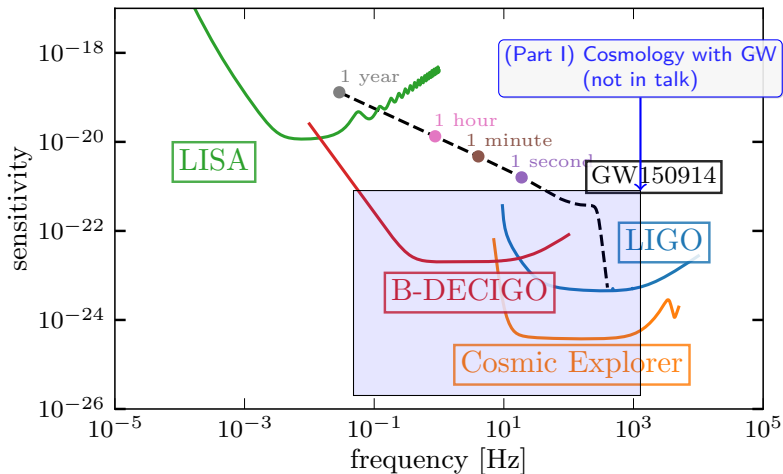
[LISA Collaboration 2017]

Current & Future Detectors

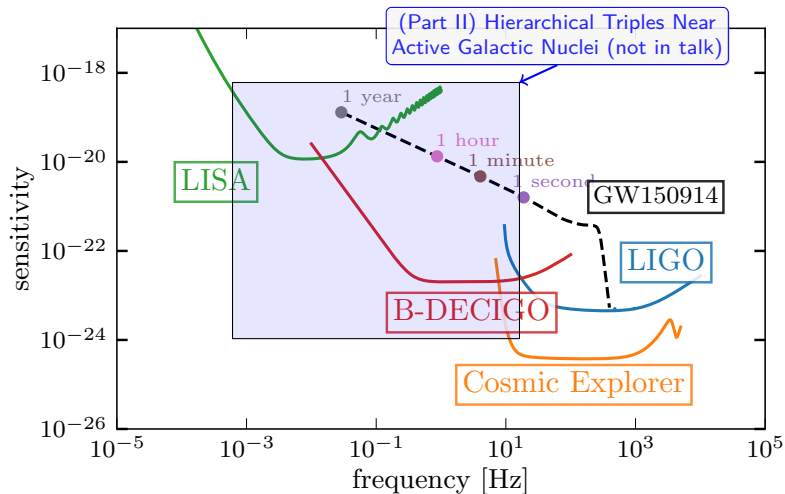


[DECIGO 2017]

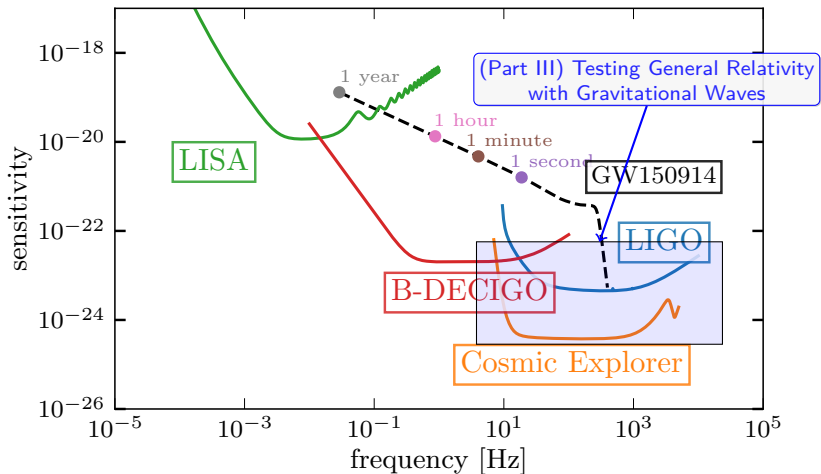
Current & Future Detectors



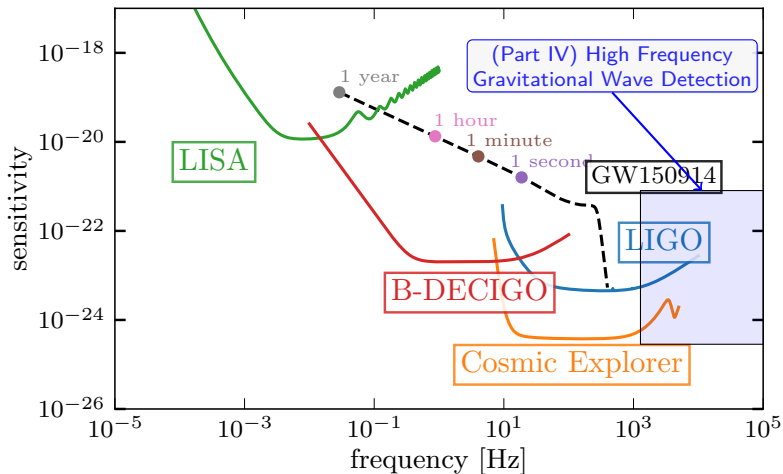
Current & Future Detectors

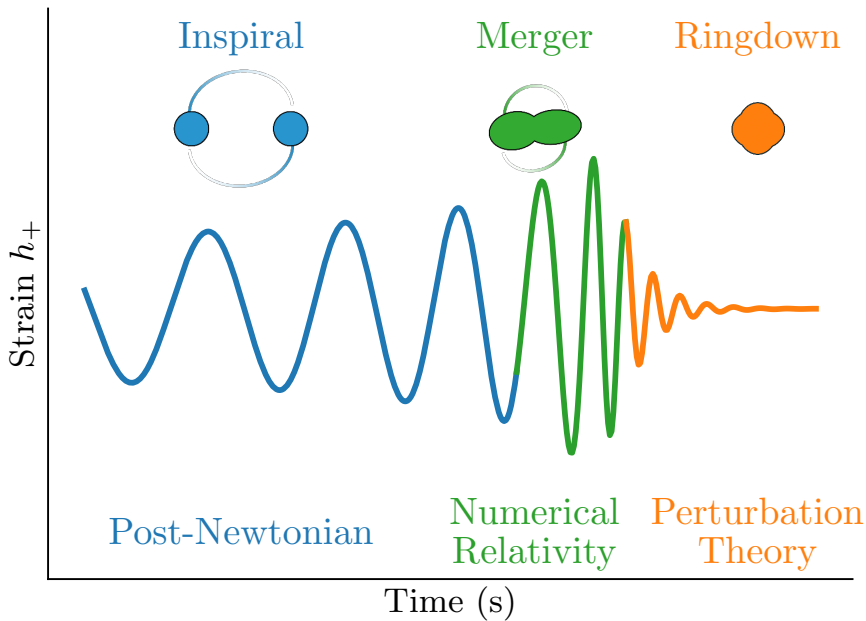


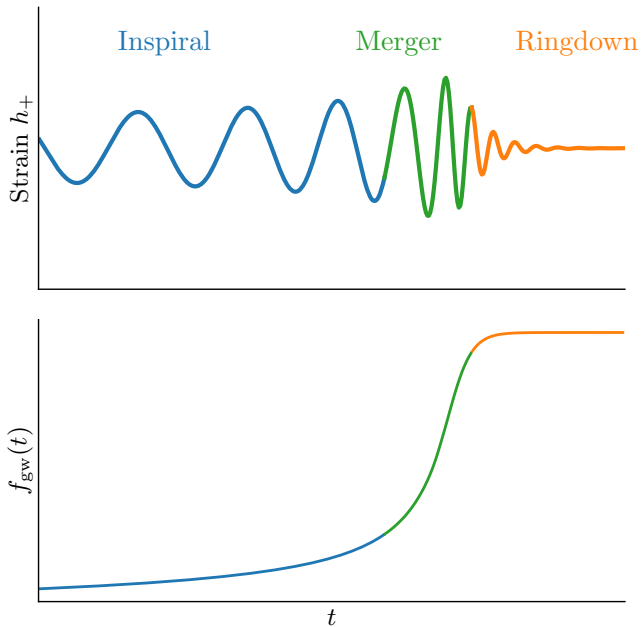
Current & Future Detectors



Current & Future Detectors

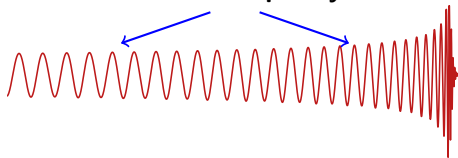






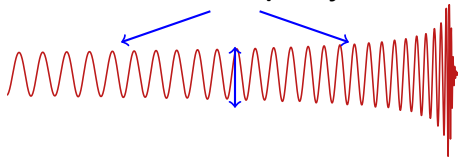
Gravitational Waves Encode Intrinsic Binary Physics

masses inferred via **frequency evolution**



Gravitational Waves Encode Intrinsic Binary Physics

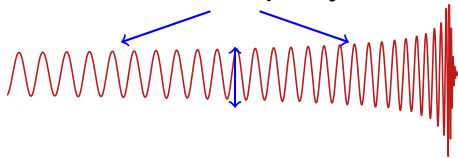
masses inferred via **frequency evolution**



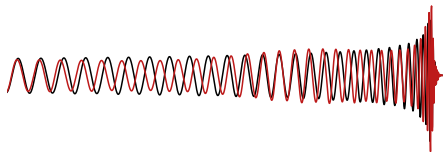
distance measured via **amplitude** and **masses**

Gravitational Waves Encode Intrinsic Binary Physics

masses inferred via **frequency evolution**



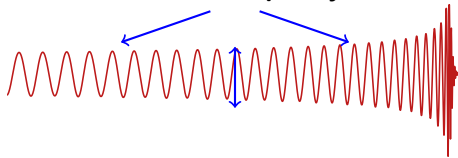
distance measured via **amplitude** and **masses**



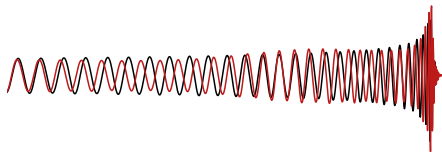
modulations of **amplitude** and **phase**
encode **spins**

Gravitational Waves Encode Intrinsic Binary Physics

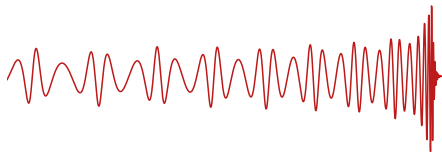
masses inferred via **frequency evolution**



distance measured via **amplitude** and **masses**

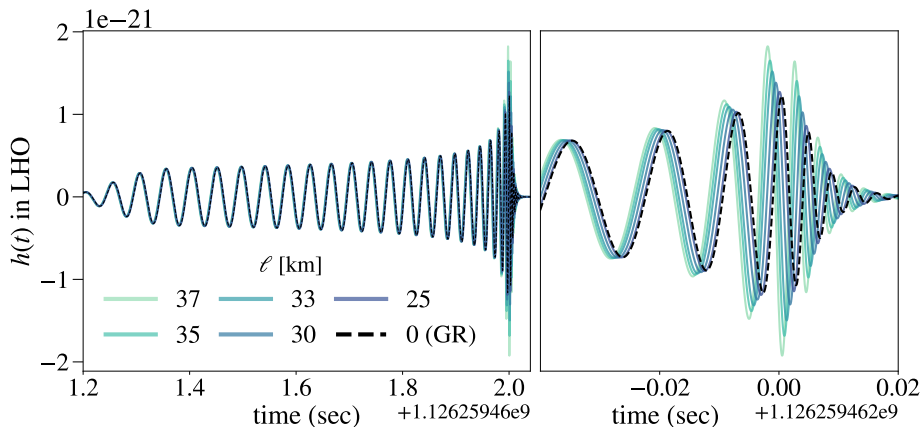


modulations of **amplitude** and **phase**
encode **spins**



eccentricity also manifests in
amplitude and **phase** modulations

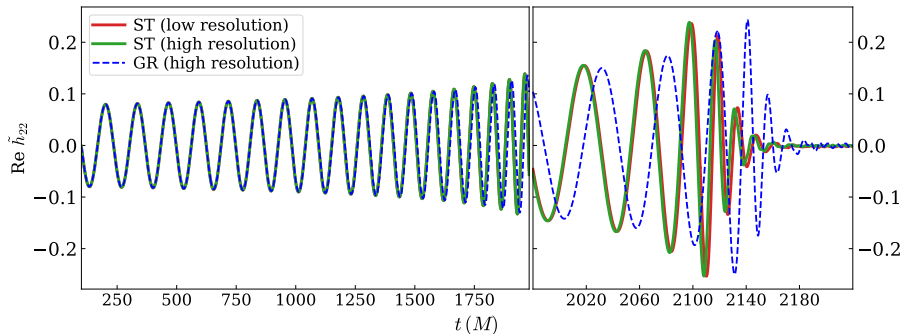
Beyond GR Waveforms



- Beyond GR simulation of dynamical Chern-Simons [[Okounkova+ 2023](#)]

Beyond GR Waveforms

$$m_{\text{BH}}^{\text{J}} = 5.7 M_{\odot}, m_{\text{NS}}^{\text{J}} = 1.5 M_{\odot}, \chi_z^{\text{BH}} = -0.19, \chi_z^{\text{NS}} = 0, \Lambda_2^{\text{GR}} = 131.1$$



- Beyond GR simulation of scalar tensor [Ma+ 2023]

Parameterized Tests of General Relativity

- Beyond GR signals are typically characterized by dephasing

$$h_{\text{bgr}}(f; \theta, \alpha) = h_{\text{gr}}(f; \theta) e^{i\Delta\Psi(f; \alpha)}$$

Parameterized Tests of General Relativity

- Beyond GR signals are typically characterized by dephasing

$$h_{\text{bgr}}(f; \theta, \alpha) = h_{\text{gr}}(f; \theta) e^{i\Delta\Psi(f; \alpha)}$$



Parameterized Tests of General Relativity

- Beyond GR signals are typically characterized by dephasing

$$h_{\text{bgr}}(f; \theta, \alpha) = h_{\text{gr}}(f; \theta) e^{i\Delta\Psi(f; \alpha)}$$



- The parameterized post-Einsteinian (ppE) framework is a common beyond GR signal morphology [Yunes+ 2009, Li+ 2011]

$$\Delta\Psi_k \propto \delta\varphi_k (\pi M f)^{(k-5)/3}$$

Parameterized Tests of General Relativity

- Beyond GR signals are typically characterized by dephasing

$$h_{\text{bgr}}(f; \theta, \alpha) = h_{\text{gr}}(f; \theta) e^{i\Delta\Psi(f; \alpha)}$$



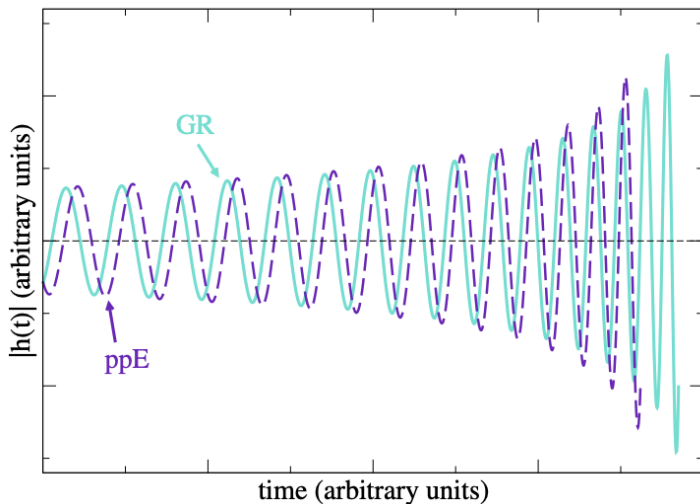
- The parameterized post-Einsteinian (ppE) framework is a common beyond GR signal morphology [Yunes+ 2009, Li+ 2011]

$$\Delta\Psi_k \propto \delta\varphi_k (\pi M f)^{(k-5)/3}$$

where each of these $\delta\varphi_k \leftrightarrow$ deviation of $(M/r)^k$ away from GR

PPE in Time Domain

- These ppE deviations in time domain look like [Carson+ 2020]



How do We Interpret Parameterized Constraints

- **Intrinsic physics** of the inspiral are encoded in the **phase** of the waveform.

How do We Interpret Parameterized Constraints

- **Intrinsic physics** of the inspiral are encoded in the **phase** of the waveform.
- The frequency chirp is related to the energy loss rate in the system by

$$\frac{df}{dt} = \frac{df}{dE} \frac{dE}{dt}$$

- $\frac{df}{dE}$ is due to modified Kepler's third law, $\frac{dE}{dt}$ due to dissipative modifications.

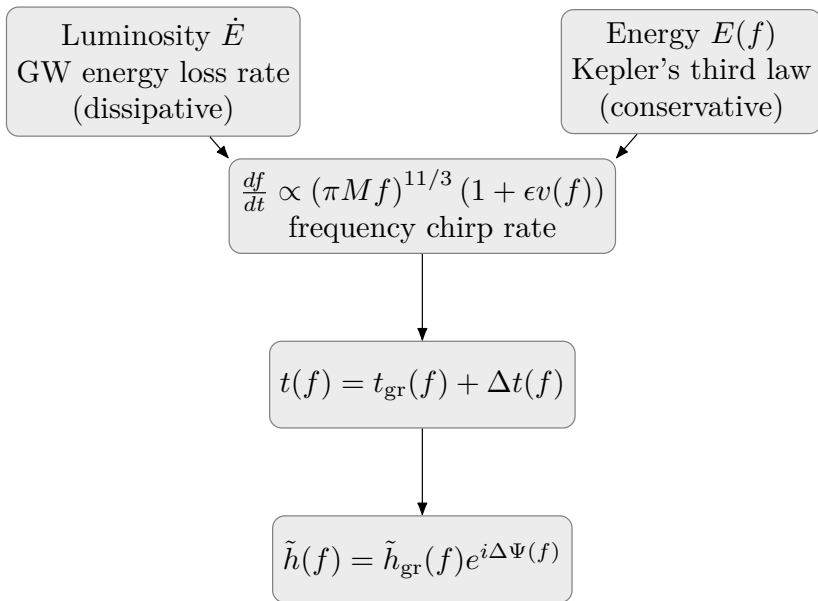
How do We Interpret Parameterized Constraints

- **Intrinsic physics** of the inspiral are encoded in the **phase** of the waveform.
- The frequency chirp is related to the energy loss rate in the system by

$$\frac{df}{dt} = \frac{df}{dE} \frac{dE}{dt}$$

- $\frac{df}{dE}$ is due to modified Kepler's third law, $\frac{dE}{dt}$ due to dissipative modifications.
- A beyond GR effect causes relative time delays $\Delta t(f)$ and then the stationary phase approximation for an adiabatic energy loss rate implies [Yunes+ 2009, Tahura+ 2019]

$$\Delta\Psi(f) = 2\pi \int df \Delta t(f)$$



Part A : Modeling the Gravitational Waves in Nonviolent Nonlocality

Possibilities for the Black Hole Unitarity Crisis

(BS: add slide with picture /or actually just informal definition possibly show page curve or do one sentence) It is proposed that the following three statements cannot be true simultaneously [Almheiri+ 2012]

- 1 Hawking radiation is a pure state.
- 2 Infalling observer feels nothing unusual at the horizon.
- 3 Hawking radiation comes from near the horizon.

(BS: say this is assumption violation examples)

Possibilities for the Black Hole Unitarity Crisis

(BS: add slide with picture /or actually just informal definition possibly show page curve or do one sentence) It is proposed that the following three statements cannot be true simultaneously [[Almheiri+ 2012](#)]

- ① Hawking radiation is a pure state.
 - Breakdown of unitary evolution.
- ② Infalling observer feels nothing unusual at the horizon.
- ③ Hawking radiation comes from near the horizon.

(BS: say this is assumption violation examples)

Possibilities for the Black Hole Unitarity Crisis

(BS: add slide with picture /or actually just informal definition possibly show page curve or do one sentence) It is proposed that the following three statements cannot be true simultaneously [Almheiri+ 2012]

- ① Hawking radiation is a pure state.
 - Breakdown of unitary evolution.
- ② Infalling observer feels nothing unusual at the horizon.
 - Infalling observer is destroyed, e.g. firewall
- ③ Hawking radiation comes from near the horizon.

(BS: say this is assumption violation examples)

Possibilities for the Black Hole Unitarity Crisis

(BS: add slide with picture /or actually just informal definition possibly show page curve or do one sentence) It is proposed that the following three statements cannot be true simultaneously [Almheiri+ 2012]

- ① Hawking radiation is a pure state.
 - Breakdown of unitary evolution.
- ② Infalling observer feels nothing unusual at the horizon.
 - Infalling observer is destroyed, e.g. firewall
- ③ Hawking radiation comes from near the horizon.
 - Horizon structure of a black hole is changed, e.g. nonviolent nonlocality.

(BS: say this is assumption violation examples)

Nonviolent Nonlocality

- Nonviolent nonlocality is a proposal by Steve Giddings that posits that the information is transferred via soft modes in the black hole atmosphere [[Giddings 2012](#), [Giddings+ 2016](#)].
- These metric fluctuations have an extent to $\sim r_S$ in contrast to the fluctuations in a firewall with extent $l_p \ll r_S$
- We have background metric, and the quantum fluctuations modify it

$$g_{\mu\nu} = g_{\mu\nu}^{\text{schw}} + n_{\mu\nu}$$

Modifications to the Metric due to Quantum Structure

- One can construct the most general metric fluctuations [Regge+ 1957], but the dominant one in ingoing Eddington-Finkelstein coordinates is [Giddings+ 2016]

$$n_{vv} = \sum_{\ell m} f^{\ell m}(v, r) Y_{\ell m}(\phi, \theta)$$

Modifications to the Metric due to Quantum Structure

- One can construct the most general metric fluctuations [Regge+ 1957], but the dominant one in ingoing Eddington-Finkelstein coordinates is [Giddings+ 2016]

$$n_{vv} = \sum_{\ell m} f^{\ell m}(v, r) Y_{\ell m}(\phi, \theta)$$

- We parameterize the random noise fluctuations as

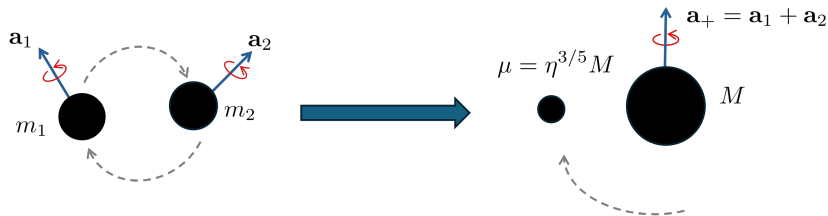
$$f^{\ell m}(v, r) = A \exp \left[-\frac{1}{2r_S^2} (r - r_S)^2 \right] n(t)$$

$$n(t) = \text{Colored gaussian noise; } \langle |n(t)| \rangle = 1$$

$$S_n(f) \propto \frac{1}{2f_Q} \exp[-|f|/f_Q]$$

Effective One-Body

- The effective one-body formalism is an analytical approach to finding the motion and gravitational waves in GR [Buonanno+ 1998, 2000, Damour 2001].

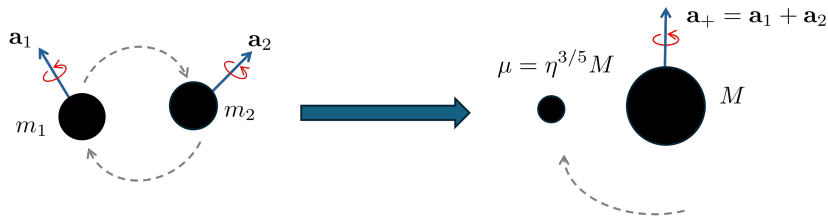


Full GR two body problem

Point particle in effective spacetime
 η -deformed Kerr

Effective One-Body

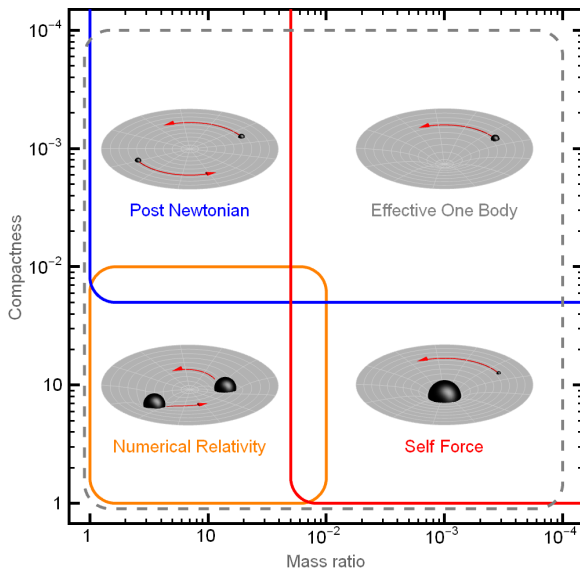
- The effective one-body formalism is an analytical approach to finding the motion and gravitational waves in GR [Buonanno+ 1998, 2000, Damour 2001].
- It resums independent information from (a) post-Newtonian theory and (b) black hole perturbation theory



Full GR two body problem

Point particle in effective spacetime
 η -deformed Kerr

Effective One-Body



Effective One-Body Details

- The effective one-body defines an effective metric

$$ds_{\text{eff}}^2 = -A(r)dt^2 + \frac{D(r)}{A(r)}dr^2 + r^2 d\Omega^2$$

where if $\eta \rightarrow 0$ it approaches the Schwarzschild metric in the nonspinning case.

- One can find that solving the mass shell condition $p_\mu p_\nu g^{\mu\nu} = -1$ yields an effective Hamiltonian

$$\hat{H}_{\text{eff}} = \sqrt{A(r) \left(1 + \frac{p_\phi^2}{r^2} + \frac{A}{D} p_r^2 \right)}$$

- The particle follows the trajectory of the real Hamiltonian

$$\hat{H}_{\text{real}} = \frac{1}{\eta} \sqrt{1 + 2\eta \left(\hat{H}_{\text{eff}} - 1 \right) + \mathcal{O}(\vec{P}_{\text{COM}}^2)}$$

where the center of mass motion is ignored.

Perturbations to Hamilton's Equations

- Nonviolent nonlocality adds perturbations to black holes which changes the mass shell condition

$$p_\mu p_\nu (g_0^{\mu\nu} + h^{\mu\nu}) = -1$$

This causes the real Hamiltonian to be modified

$$\hat{H}_{\text{real}} = \hat{H}_{\text{real}}^0 + h_{\nu\nu}^{\ell m} \Delta \hat{H}_{\ell m}^{\text{real}}.$$

(BS: consider combining or simplifying htis part. to reduce complexity)

Perturbations to Hamilton's Equations

- Nonviolent nonlocality adds perturbations to black holes which changes the mass shell condition

$$p_\mu p_\nu (g_0^{\mu\nu} + h^{\mu\nu}) = -1$$

This causes the real Hamiltonian to be modified

$$\hat{H}_{\text{real}} = \hat{H}_{\text{real}}^0 + h_{\nu\nu}^{\ell m} \Delta \hat{H}_{\ell m}^{\text{real}}.$$

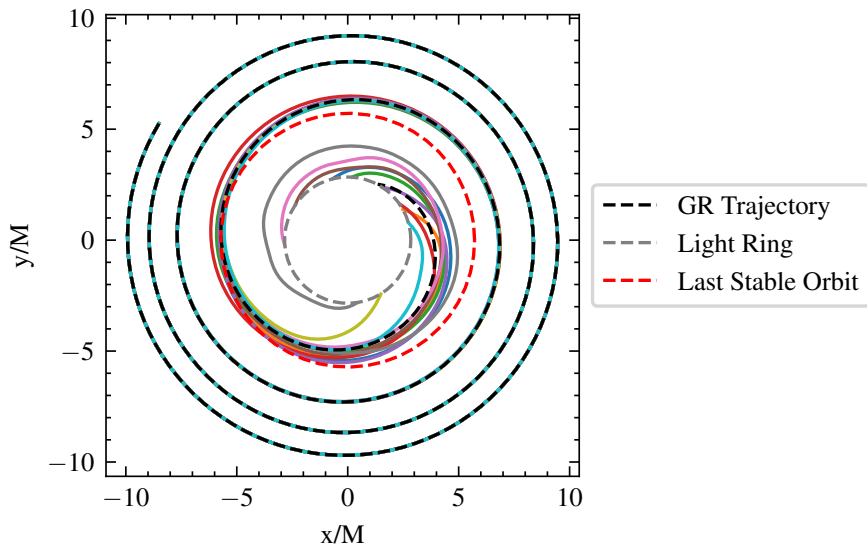
- In the end, we integrate Hamilton's equations which have the form

$$\begin{aligned} \frac{dq_i}{dt} &= \frac{\partial \hat{H}_{\text{real}}}{\partial p_i}, \\ \frac{dp_i}{dt} &= -\frac{\partial \hat{H}_{\text{real}}}{\partial q_i} + \mathcal{F}_i^{\text{rad}}, \end{aligned}$$

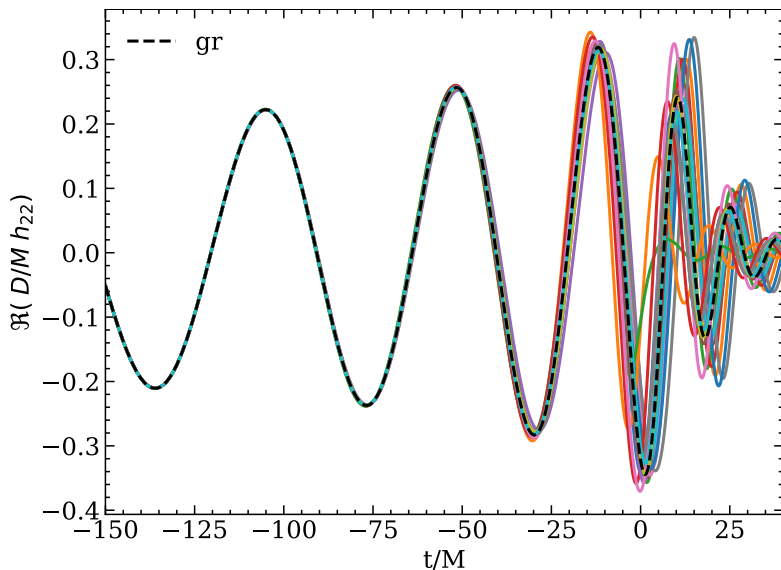
where $\mathcal{F}_i^{\text{rad}}$ are the nonconservative forces from radiation reaction.

(BS: consider combining or simplifying htis part. to reduce complexity)

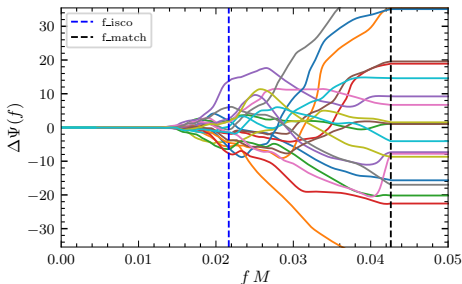
Inspiral with Quantum Fluctuations



Change in Gravitational Wave Strain

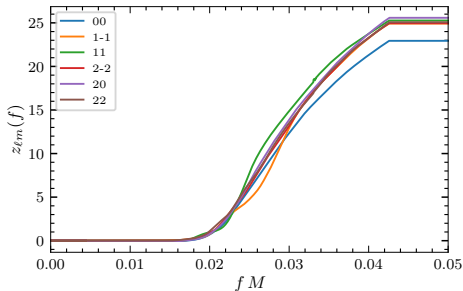


Random Dephasing in Frequency Domain



- Principal component analysis yields deviation that looks like $\Delta\Psi(f) = \zeta z(f)$
- $\zeta \sim \mathcal{N}(0, A_t)$ random variable for each event

PCA



Frequency Domain Dephasing Model

- From the figure before, we saw that the random metric fluctuations produce dephasing which has a lot of structure.
- We use a principal component analysis to model the dephasing in a simple manner.

Frequency Domain Dephasing Model

- From the figure before, we saw that the random metric fluctuations produce dephasing which has a lot of structure.
- We use a principal component analysis to model the dephasing in a simple manner.
- The mean deviation and covariance matrix are defined as

$$\mu \equiv \langle \Delta\Psi(f) \rangle$$

$$\Sigma(f, f') = \langle (\Delta\Psi(f) - \mu(f)) (\Delta\Psi(f') - \mu(f')) \rangle$$

Frequency Domain Dephasing Model

- From the figure before, we saw that the random metric fluctuations produce dephasing which has a lot of structure.
- We use a principal component analysis to model the dephasing in a simple manner.
- The mean deviation and covariance matrix are defined as

$$\mu \equiv \langle \Delta\Psi(f) \rangle$$

$$\Sigma(f, f') = \langle (\Delta\Psi(f) - \mu(f)) (\Delta\Psi(f') - \mu(f')) \rangle$$

- The principal component analysis finds optimal eigenvectors

$$\Sigma(f, f') \approx A^2 z(f)z(f'),$$

Hierarchical Tests of GR

- What we have shown is that nonviolent nonlocality predicts stochastic deviations to the phase $\Delta\Psi(f) = \zeta z(f)$.

$$\zeta \sim \mathcal{N}(0, A)$$

- This is of the same form as the hierarchical tests of GR [Isi+ 2019] which are published in the LIGO papers [LIGO+ 2021].

$$\delta\phi_k \sim \mathcal{N}(\mu_k, \sigma_k)$$

where these are the deformation coefficients.

(BS: consider combining or simplifying this part. to reduce complexity)

Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_l, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

where $(. | .)$ is the standard noise weighted inner product.

Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_l, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

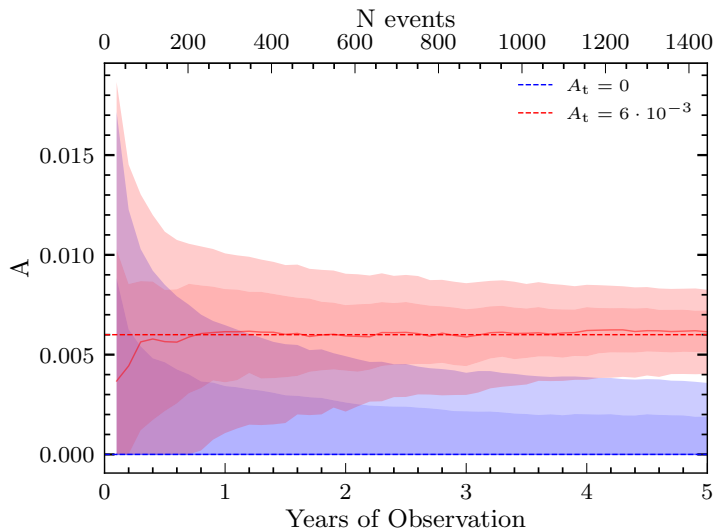
where $(. | .)$ is the standard noise weighted inner product.

- From this we calculate the marginalized likelihood $p(d_a | \zeta)$ for each event a . The likelihood for the hyper parameters is

$$p(d | \mu, \sigma) = \int d\zeta p(d | \zeta) p(\zeta | \mu, \sigma)$$

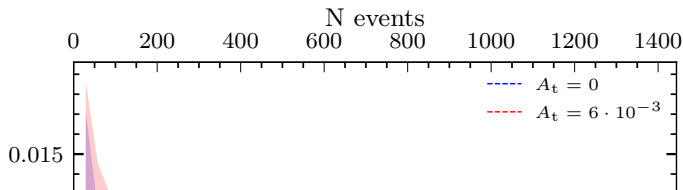
- From this, we compute the posterior on $A \equiv \sigma$ by combining many events d_a .

Constraints on Nonviolent Nonlocality with LIGO

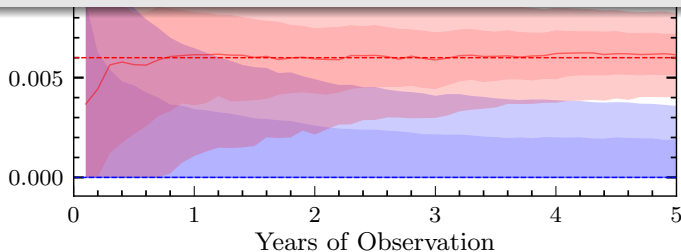


[backup slide] (BS)

Constraints on Nonviolent Nonlocality with LIGO



Metric fluctuations can be constrained to be $A \lesssim 6 \times 10^{-6}$ with 5 years of O3 data.



[backup slide] (BS)

Part B : Geometric Description of Tests of GR

Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$

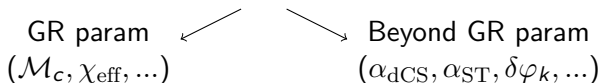
Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$



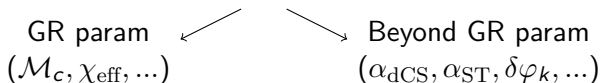
Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$



- The true beyond GR phase modification is

$$\Delta\Psi_{\text{bgr}}(f, \alpha) = \alpha\psi_{\alpha}(f)$$

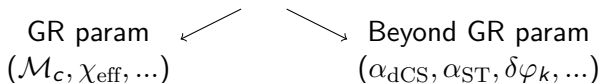
Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$



- The true beyond GR phase modification is

$$\Delta\Psi_{\text{bgr}}(f, \alpha) = \alpha\psi_{\alpha}(f)$$

- A ppE test searches for power law deviations

$$\Delta\Psi_k = \delta\varphi_k (\pi\mathcal{M}_c f)^{(k-5)/3}$$

Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$

- The true beyond GR phase modification is

$$\Delta\Psi_{\text{bgr}}(f, \alpha) = \alpha\psi_{\alpha}(f)$$

- A ppE test searches for power law deviations

$$\Delta\Psi_k = \delta\varphi_k (\pi\mathcal{M}_c f)^{(k-5)/3}$$

- ① How **degenerate with GR** parameters is $\Delta\Psi_{\text{bgr}}$?

Introduction

(BS: title fox)

- Gravitational wave data is a combination of signal and noise

$$d = s + n$$

$$s = h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) = h_{\text{gr}}(f; \theta_{\text{tr}}) e^{i\Delta\Psi_{\text{bgr}}}$$

- The true beyond GR phase modification is

$$\Delta\Psi_{\text{bgr}}(f, \alpha) = \alpha\psi_{\alpha}(f)$$

- A ppE test searches for power law deviations

$$\Delta\Psi_k = \delta\varphi_k (\pi\mathcal{M}_c f)^{(k-5)/3}$$

- How **degenerate with GR** parameters is $\Delta\Psi_{\text{bgr}}$?
- How **accurately** ppE $\Delta\Psi_k$ captures the true deviation $\Delta\Psi_{\text{bgr}}$?

Residual Signal

(BS: just put d instead of $sbgr$)

- Doing PE on $h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}})$ with a GR waveform $h_{\text{gr}}(\theta)$, the residual signal

$$\begin{aligned}\Delta h &= h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}}) - h_{\text{gr}}(\theta_{\text{tr}}) \\ &\approx i\alpha_{\text{tr}}\psi_{\alpha}h_{\text{gr}}(\theta_{\text{tr}})\end{aligned}$$

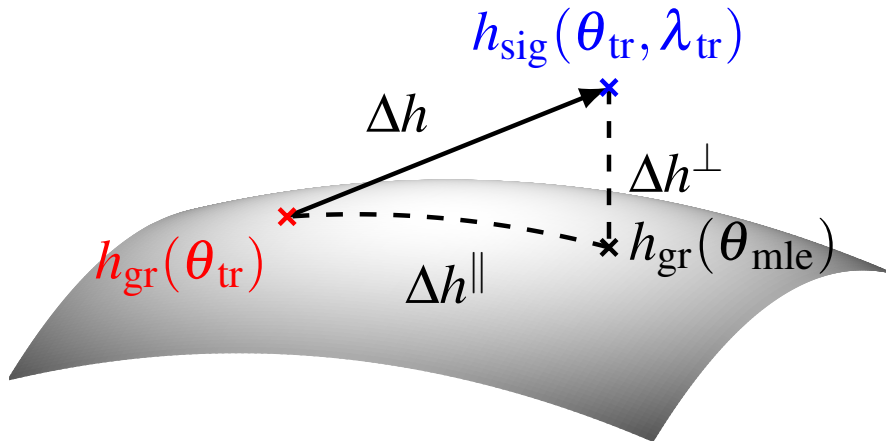
- GR parameters are biased due to systematic error [[Cutler+ 2007](#)]

$$\theta_{\text{meas}} \rightarrow \theta_{\text{tr}} + \Delta\theta_{\text{bias}}$$

- The residual signal is perpendicular part of the waveform

$$\Delta h_{\perp\text{gr}} = \Delta h - \Delta\theta_{\text{bias}}^i \partial_i h_{\text{gr}}$$

(BS: For example if you dont model spins, q and mc will be biased)



Must remake with $\lambda - > \alpha$

(E

Perpendicular SNR and Bayes Factors

- Given an injected signal $h_{\text{sig}}(\theta_{\text{tr}}, \alpha_{\text{tr}})$, we compute the evidence for both GR and bGR [$p(\text{GR}|s_{\text{bGR}})$ and $p(\text{bGR}|s_{\text{bGR}})$]
- The Bayes factor compares the evidence for a beyond GR theory in the data

$$\mathcal{O}_{\text{GR}}^{\text{bGR}} \equiv \frac{p(\text{bGR}|s_{\text{bGR}})}{p(\text{GR}|s_{\text{bGR}})}$$

- The Bayes factor behaves like [\[Vallisneri 2009, 2013\]](#)

$$\log \mathcal{O}_{\text{GR}}^{\text{bGR}}|_{s_{\text{bGR}}} \sim \frac{1}{2}\rho_{\perp}^2 + \rho_{\perp}x + \frac{1}{2}x^2 \quad (\text{bGR injection})$$

$$\log \mathcal{O}_{\text{GR}}^{\text{bGR}}|_{s_{\text{GR}}} \sim \frac{1}{2}x^2 \quad (\text{GR injection})$$

where $\rho_{\perp} = \|s_{\text{bGR}}^{\perp}\|$ and x is random unit normal variable.

Bayes Factor for Parameterized Test

- How **accurately** does $\text{ppE } \Delta\Psi_k$ captures the true deviation $\Delta\Psi_{\text{bgr}}$?

Bayes Factor for Parameterized Test

- How **accurately** does ppE $\Delta\Psi_k$ captures the true deviation $\Delta\Psi_{\text{bgr}}$?
- The Bayes factor with an incorrect ppE model is

$$\log \mathcal{O}_{\text{GR}}^{\text{ppE}} | s_{\text{bGR}} \sim \frac{1}{2} \left(\rho_{\perp}^{\text{ppE}} \right)^2 + x \rho_{\perp}^{\text{ppE}} + \frac{1}{2} x^2$$

where the captured SNR is

$$\rho_{\perp}^{\text{ppE}} = \text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) \rho_{\perp}$$

- The *fitting factor* describes how much of the bGR signal is captured by the ppE model

$$\text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) = \frac{\left(\Delta h_{\text{bGR}}^{\perp} | \Delta h_{\text{ppE}}^{\perp} \right)}{\| \Delta h_{\text{bGR}}^{\perp} \| \| \Delta h_{\text{ppE}}^{\perp} \|}$$

Bayes Factor for Parameterized Test

- How **accurately** does ppE $\Delta\Psi_k$ captures the true deviation $\Delta\Psi_{\text{bgr}}$?
- The Bayes factor with an incorrect ppE model is

$$\log \mathcal{O}_{\text{GR}}^{\text{ppE}}|_{s_{\text{bGR}}} \sim \frac{1}{2} \left(\rho_{\perp}^{\text{ppE}} \right)^2 + x \rho_{\perp}^{\text{ppE}} + \frac{1}{2} x^2$$

where the captured SNR is

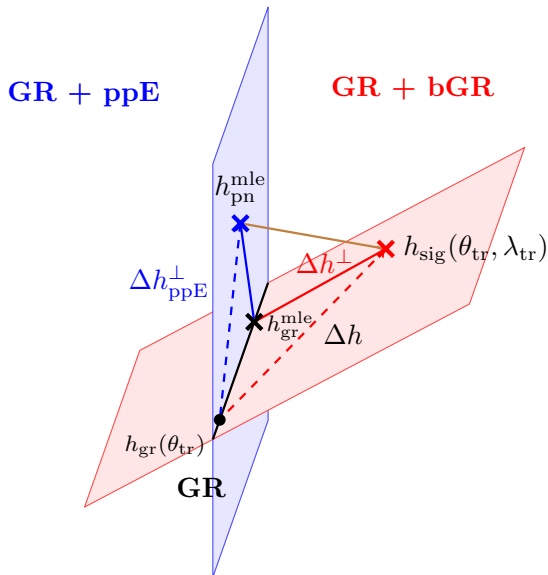
$$\rho_{\perp}^{\text{ppE}} = \text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) \rho_{\perp}$$

- The *fitting factor* describes how much of the bGR signal is captured by the ppE model

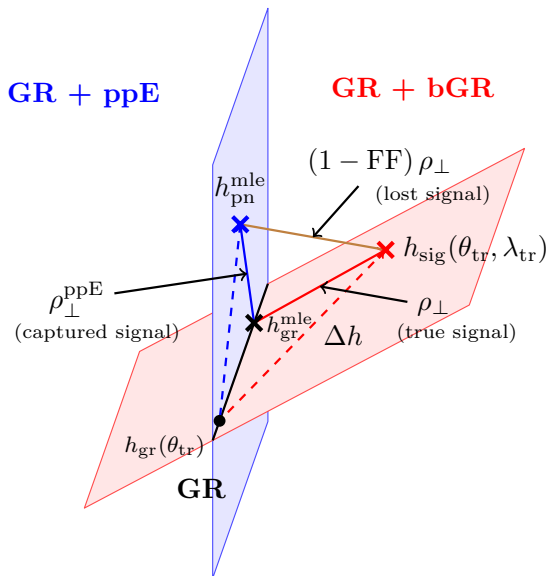
$$\text{FF}(\Delta h_{\text{bGR}}^{\perp}, \Delta h_{\text{ppE}}^{\perp}) = \frac{\left(\Delta h_{\text{bGR}}^{\perp} | \Delta h_{\text{ppE}}^{\perp} \right)}{\| \Delta h_{\text{bGR}}^{\perp} \| \| \Delta h_{\text{ppE}}^{\perp} \|}$$

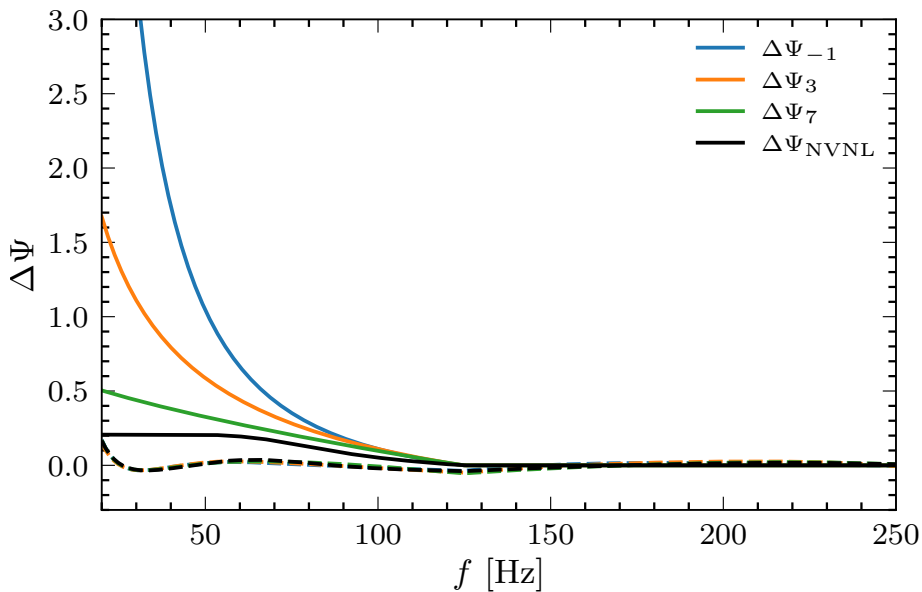
- We have found that that the sensitivity loss is very small ($1 - \text{FF} \ll 1$) for some non-PN theories [Seymour+ 2024].
 - $\Delta\Psi_{\text{NVNL}} \sim e^{-f^{-1}}$ has essential singularity at $f = 0 \implies$ no ppE power

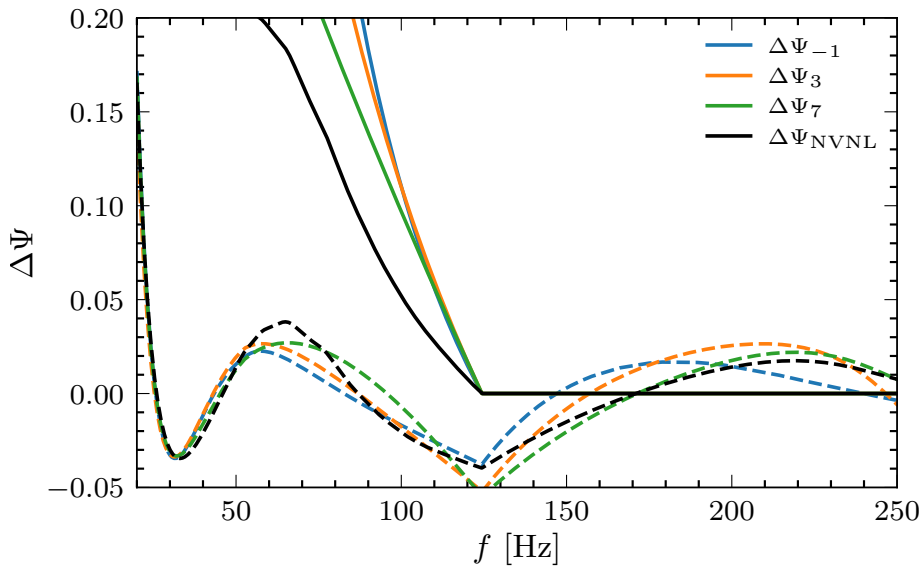
Visualization of Test



Visualization of Test

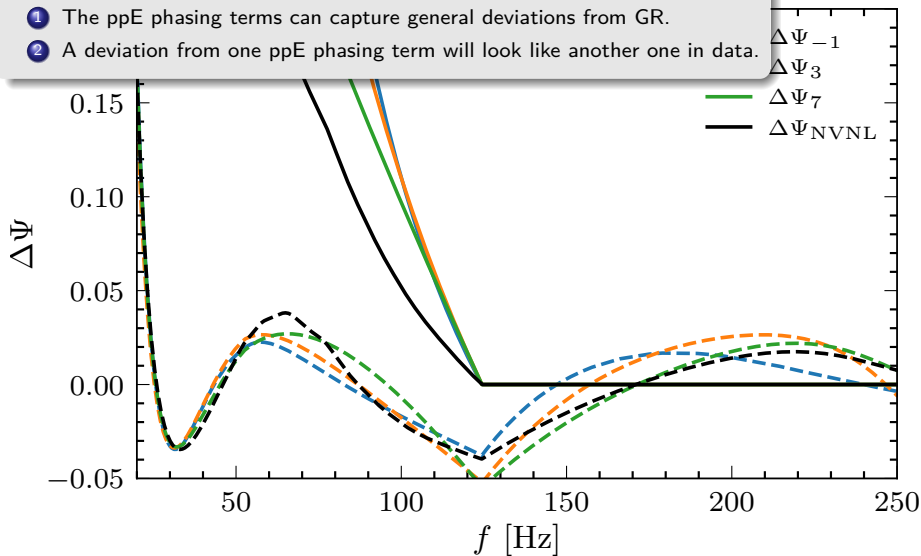




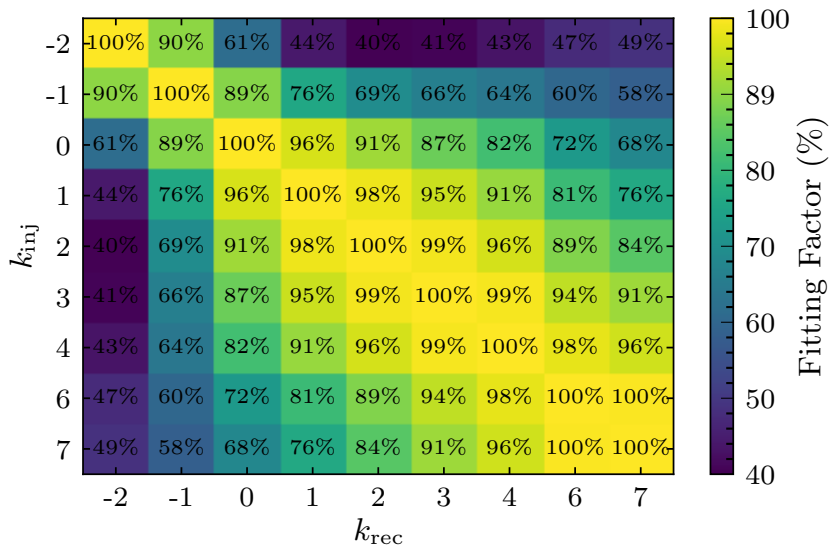


Monotonic phase deviations have some universal features in how they deviate from GR.

- ① The ppE phasing terms can capture general deviations from GR.
- ② A deviation from one ppE phasing term will look like another one in data.



Degeneracy of Multiparameter ppE



Singular Value Decomposition Approach

- Since the ppE parameter tests are degenerate, we need to identify common modes of the waveform
- We generalize the singular value decomposition [Pai+ 2013] to identify nondegenerate deviations from GR

Singular Value Decomposition Approach

- Since the **ppE** parameter tests are degenerate, we need to identify common modes of the waveform
- We generalize the singular value decomposition [Pai+ 2013] to identify nondegenerate deviations from GR
- The **singular value decomposition** finds the common features by identifying common features Δh_α [Tiglio+ 2022]

$$C(\Delta h_\alpha) = \sum_a \|\Delta h_a - \mathcal{P}_n \Delta h_a\|^2$$

where the projection $\mathcal{P}_n$ projects to an orthonormal SVD basis

$$\mathcal{P}_n \Delta h_a = \sum_\alpha (\Delta h_a | \Delta h_\alpha) \Delta h_\alpha$$

$$(\Delta h_\alpha | \Delta h_\beta) = \delta_{\alpha\beta}$$

Singular Value Decomposition Approach

- Since the ppE parameter tests are degenerate, we need to identify common modes of the waveform
- We generalize the singular value decomposition [Pai+ 2013] to identify nondegenerate deviations from GR
- The **singular value decomposition** finds the common features by identifying common features Δh_α [Tiglio+ 2022]

$$C(\Delta h_\alpha) = \sum_a \|\Delta h_a - \mathcal{P}_n \Delta h_a\|^2$$

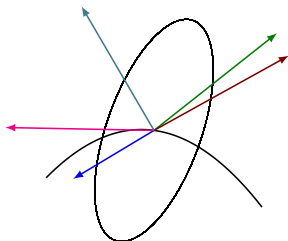
where the projection $\mathcal{P}_n$ projects to an orthonormal SVD basis

$$\mathcal{P}_n \Delta h_a = \sum_\alpha (\Delta h_a | \Delta h_\alpha) \Delta h_\alpha$$

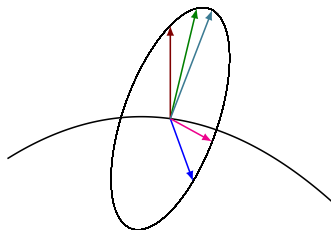
$$(\Delta h_\alpha | \Delta h_\beta) = \delta_{\alpha\beta}$$

- Goal is to find $n_{\text{svd}} \ll n_{\text{ppE}}$ but still fit signal well

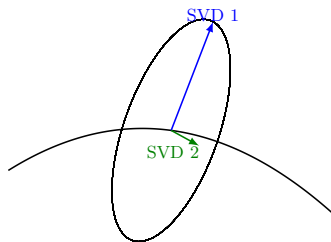
Visualization of Multiparameter SVD



Step 1: compute Δh_{ppE}

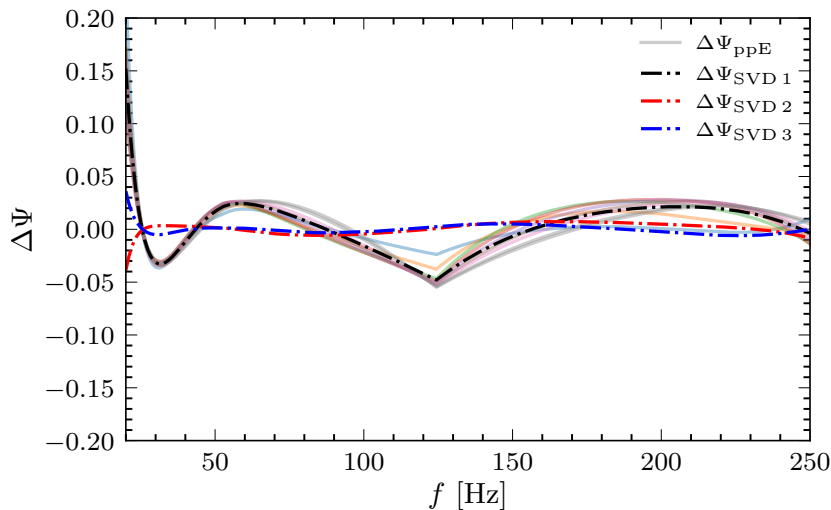


Step 2: $\Delta h_{ppE}^\perp$ computed by projecting terms parallel to $\partial_{\theta^i} h_{gr}$



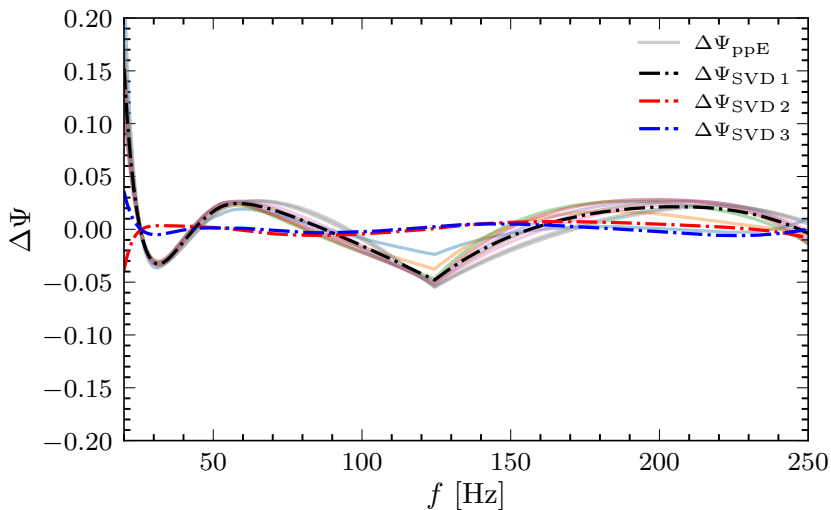
Step 3: Compute $\Delta h_{SVD}^\perp$ with the singular value decomposition

Multiparameter SVD Example for GW150914



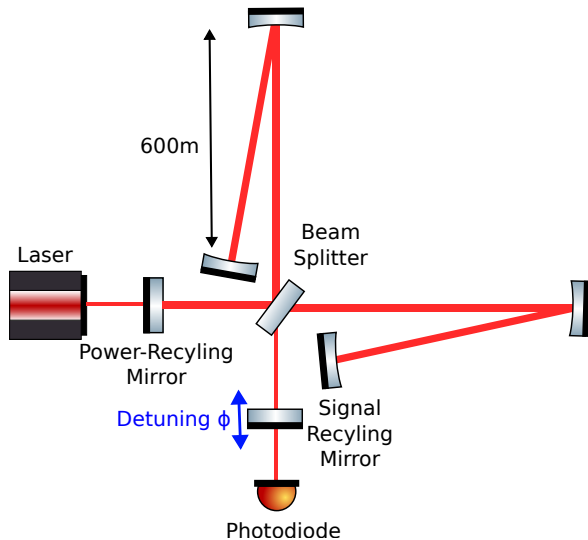
These SVD waveform modes are orthogonal so that

$$(i\Delta\Psi_{\text{SVD } a}h_{\text{gr}}|i\Delta\Psi_{\text{SVD } b}h_{\text{gr}}) = s_a^2\delta_{ab}$$



Part C : High Frequency Gravitational Wave Detection

GEO600 Optical Layout



How Detectors Measure Gravitational Waves

- The light at the photodiode is a mixture of signal and noise.

$$i(f) = T_h(f)h(f) + \sum_i T_i(f)n_i(f)$$

How Detectors Measure Gravitational Waves

- The light at the photodiode is a mixture of signal and noise.

$$i(f) = T_h(f)h(f) + \sum_i T_i(f)n_i(f)$$

- The PSD is then equal to

$$S_h(f) = \frac{1}{|T_h(f)|^2} \sum_i |T_i(f)|^2 S_j(f)$$

How Detectors Measure Gravitational Waves

- The light at the photodiode is a mixture of signal and noise.

$$i(f) = T_h(f)h(f) + \sum_i T_i(f)n_i(f)$$

- The PSD is then equal to

$$S_h(f) = \frac{1}{|T_h(f)|^2} \sum_i |T_i(f)|^2 S_j(f)$$

- Can we modify the **optimal frequency** that we are sensitive at by detuning the **location** of the signal recycling mirror?

How Detectors Measure Gravitational Waves

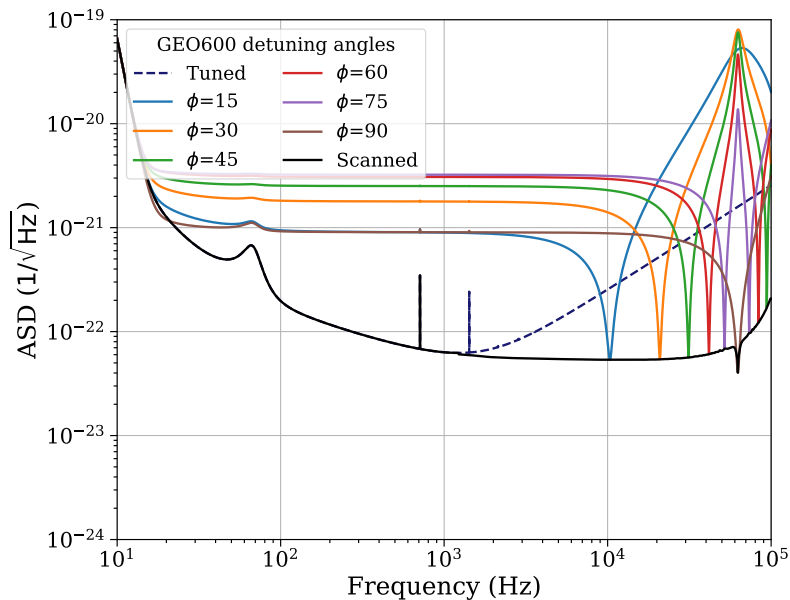
- The light at the photodiode is a mixture of signal and noise.

$$i(f) = T_h(f)h(f) + \sum_i T_i(f)n_i(f)$$

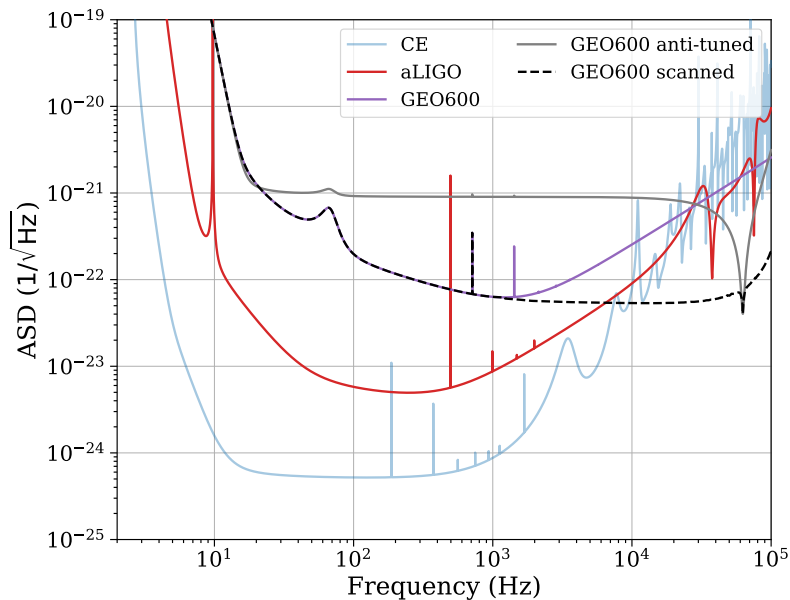
- The PSD is then equal to

$$S_h(f) = \frac{1}{|T_h(f)|^2} \sum_i |T_i(f)|^2 S_j(f)$$

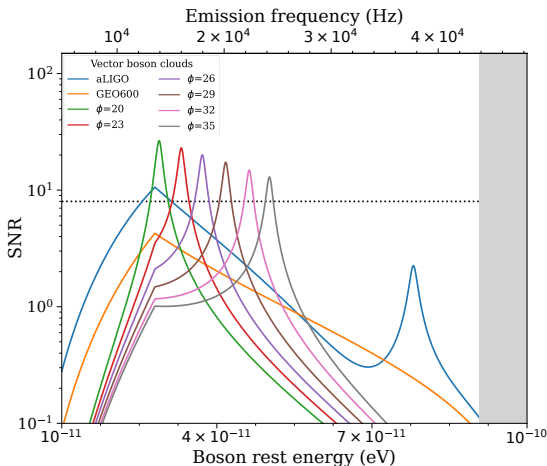
- Can we modify the **optimal frequency** that we are sensitive at by detuning the **location** of the signal recycling mirror?
- $T_h(f; \phi_{\text{srm}})$ can be adjusted to search for gravitational waves at high frequencies [Meers 1988, Mizuno+ 1993].



[backup-slide]



[backup-slide]



- GEO600 detector improves the detection prospects on boson clouds ($M = 0.3 M_{\odot}$, $\chi = 0.7$ using model from [Isi+ 2018])
- Quasircular sub-solar mass mergers $10^{-3} M_{\odot} \lesssim M \lesssim 10^{-1} M_{\odot}$ have no improvement in sensitivity.

[backup-slide]

Conclusions

- ① Studied the prospects for detecting nonviolent nonlocality in LIGO
 - Modeled the waveform in the effective-one-body formalism
 - Showed that stochastic deviations to phase are expected
 - Stacked together many events to estimate constraints in LIGO
- ② Identified geometrical meaning of tests of GR
 - Explained degeneracies and significance by geometrical framework
 - Characterized systematic error of using parameterized models
 - Used singular value decomposition to identify common modes
- ③ Modeled the effects of high frequency sensitivity in GEO600 by modulating the signal recycling cavity location.

(BS: need to improve thie conclusions slide!!) (BS: Consider adding HF Geo slides) (BS: Need to have thank you slides!!)

Appendix/Backup Slides

Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_I, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

where $(.|.)$ is the standard noise weighted inner product.

Fisher Analysis

- We inject $\theta = (\zeta, \mathcal{M}_c, q, D_I, \iota, \psi, \alpha, \delta)$ from realistic astrophysical populations and compute the estimated variance with the Fisher matrix

$$\Gamma_{ij} = (\partial_{\theta_i} h | \partial_{\theta_j} h)$$

where $(. | .)$ is the standard noise weighted inner product.

- From this we calculate the marginalized likelihood $p(d_a | \zeta)$ for each event a . The likelihood for the hyper parameters is

$$p(d | \mu, \sigma) = \int d\zeta p(d | \zeta) p(\zeta | \mu, \sigma)$$

- From this, we compute the posterior on the hyper parameters

$$p(\{d\} | \mu, \sigma) = \prod_a p(d_a | \mu, \sigma)$$

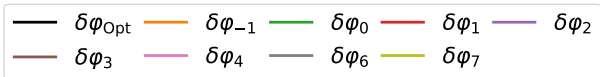
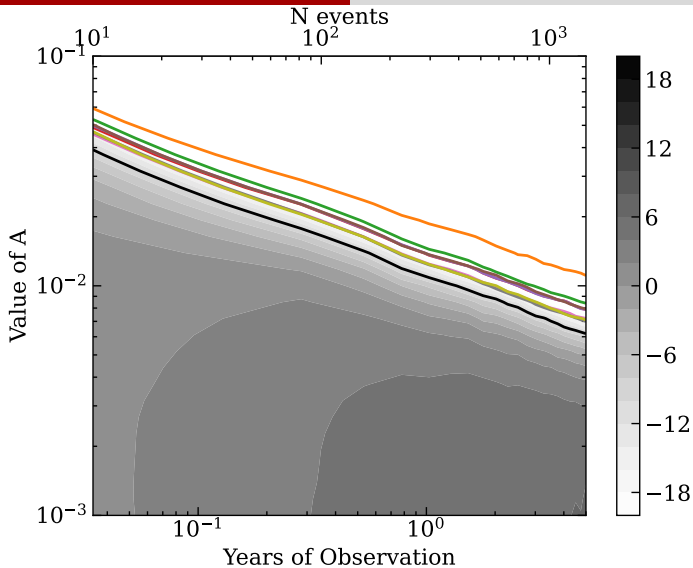
Bayes Factor

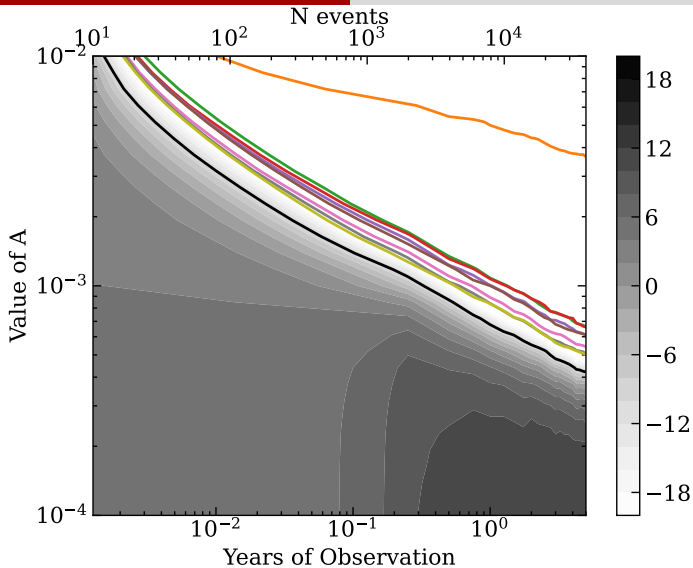
- To compare the consistency with GR, we use the log Bayes factor

$$\mathcal{B}_{\text{GR}}^{\text{NVNL}} = \log \left(\frac{p(\{d\} | M_{\text{NVNL}})}{p(\{d\} | M_{\text{GR}})} \right)$$

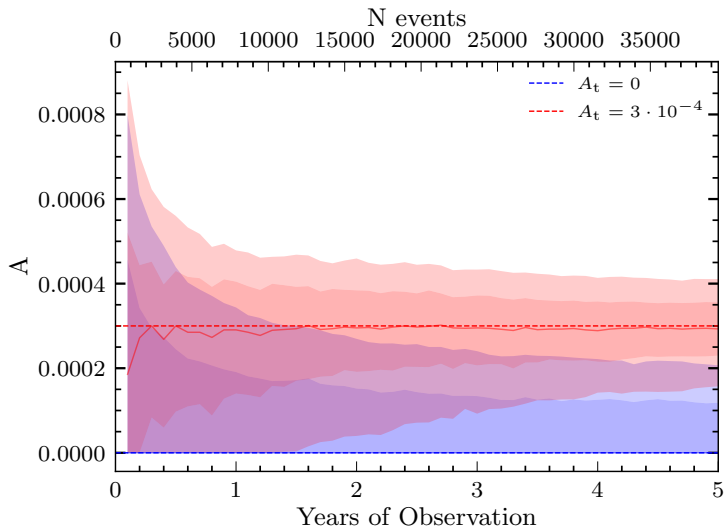
where M_{GR} and M_{NVNL} are the models.

- This is a scalar statistic which quantifies whether GR is preferred (positive) or NVNL (negative).

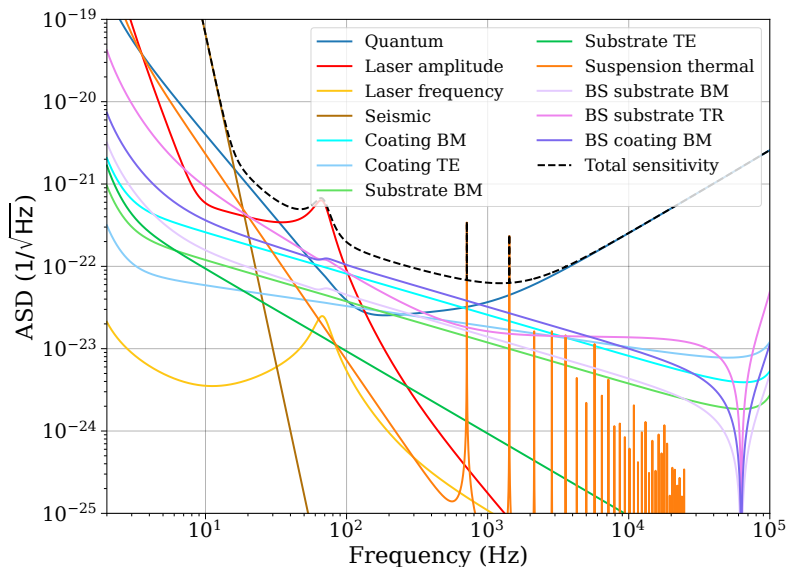




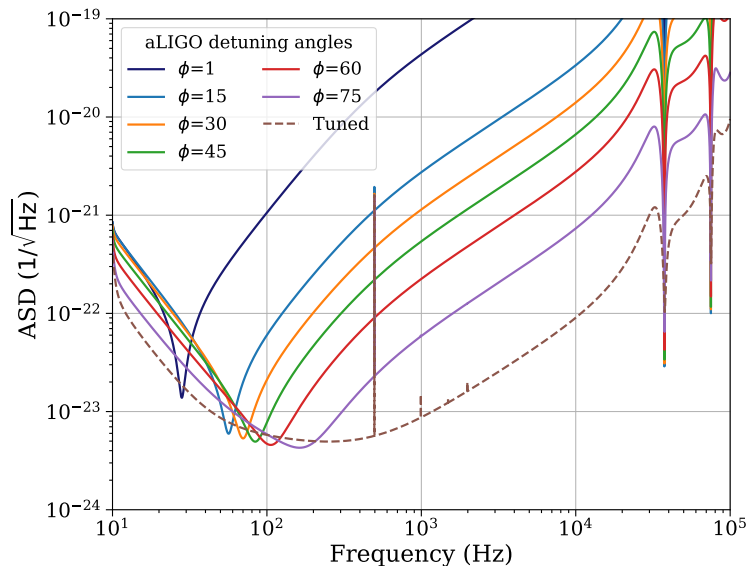
Constraints on Nonviolent Nonlocality with CE



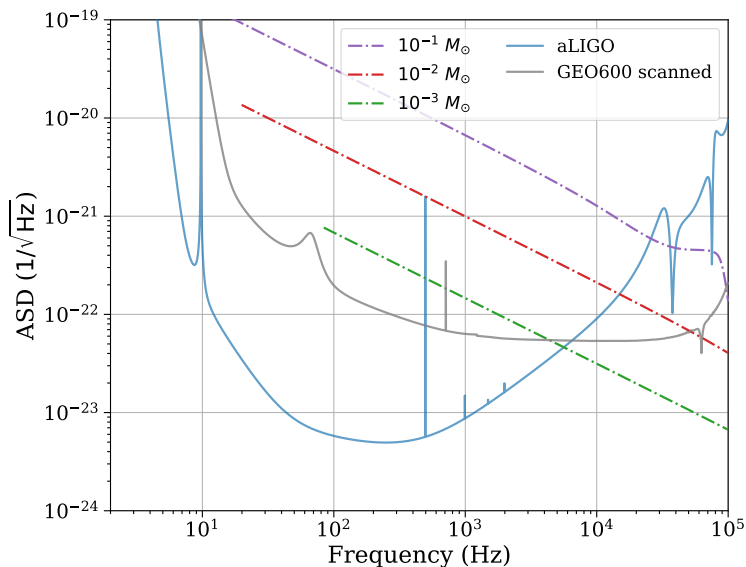
GEO600 Noise Budget



Detuning LIGO



Sub Solar Mass Detuned GEO600



Boson Clouds Full Plot

